2.0 GIS DATA TYPES AND MODELS

Spatial data are computer representations of spatial features of the real-world Representation of Real World. A modeling language for a GIS database is a spatial data model. A spatial database holds a digital representation of the real world. Among spatial data models, we can distinguish two major types, field and object-based models.

❖ **Field-based models** consider spatial phenomena to be of a continuous nature where in every point in space a value of the field can be determined. Examples of such phenomena are temperature, barometric pressure, or elevation.

❖ **Object-based models** consider space to be populated by well distinguishable, discrete, bounded objects with the space between objects potentially being empty.

Examples include parcels, buildings, trees, among others It is therefore common to see geographical phenomenon of spatial features in terms of field, objects and boundary- boundary simply defining extent of interest.

i.e Geographic phenomenon= Field, Object, Boundary

The primitives that are used to represent features in the object view comprise point, line, area/polygon, and volume items. It is often called the vector-based approach.

In the field view, the part of space relevant to us is divided up completely into regular or irregular tessellations-usually squares, triangles, or cubes-that define a two or three dimensional raster or grid. The more common name for the field view approach is the raster-based approach.

2.1 GIS Data Types

A GIS deals with spatial data (e.g., parcels, rivers, wells,), their attributes and characteristics (e.g., location, area, length, name, depth …) and the relationships between the objects (e.g., a parcel boundary follows a river, a well is located in a certain parcel …).

The objects are stored in the database with geometric primitives (volumes, areas, lines, and points) and the relationships between them (topology). Spatial data have the following characteristics:

Table 2.1: Characteristics of spatial data

Spatial reference (geographic location, coordinates)	where?
Attributes (non-spatial)	what?
Spatial relationships (topology, metric, order)	in what relationship?
Temporal component (different concepts of time)	when?

Source: Kainz, Wolfgang (2004)

It is common to equate GIS Data with three “triplex” characteristics-description, reference and time). Description is the attribute, reference is the spatial reference/data, and time as the reference and description may change over time.

The basic data type in a GIS reflects traditional data found on a map, and currently digitally in GIS or spatial databases. Accordingly, GIS technology utilizes two basic types of data. These are: spatial data (location and geometry), attribute data (descriptions) and spatial relationships.

Spatial data: describes the absolute and relative location of geographic features.

Attribute data: describes characteristics of the spatial features. These characteristics can be quantitative and/or qualitative in nature. Attribute data is often referred to as tabular data.

The coordinate location of a forestry stand would be spatial data, while the characteristics of that forestry stand, e.g. cover group, dominant species, crown closure, height, etc., would be attribute data.

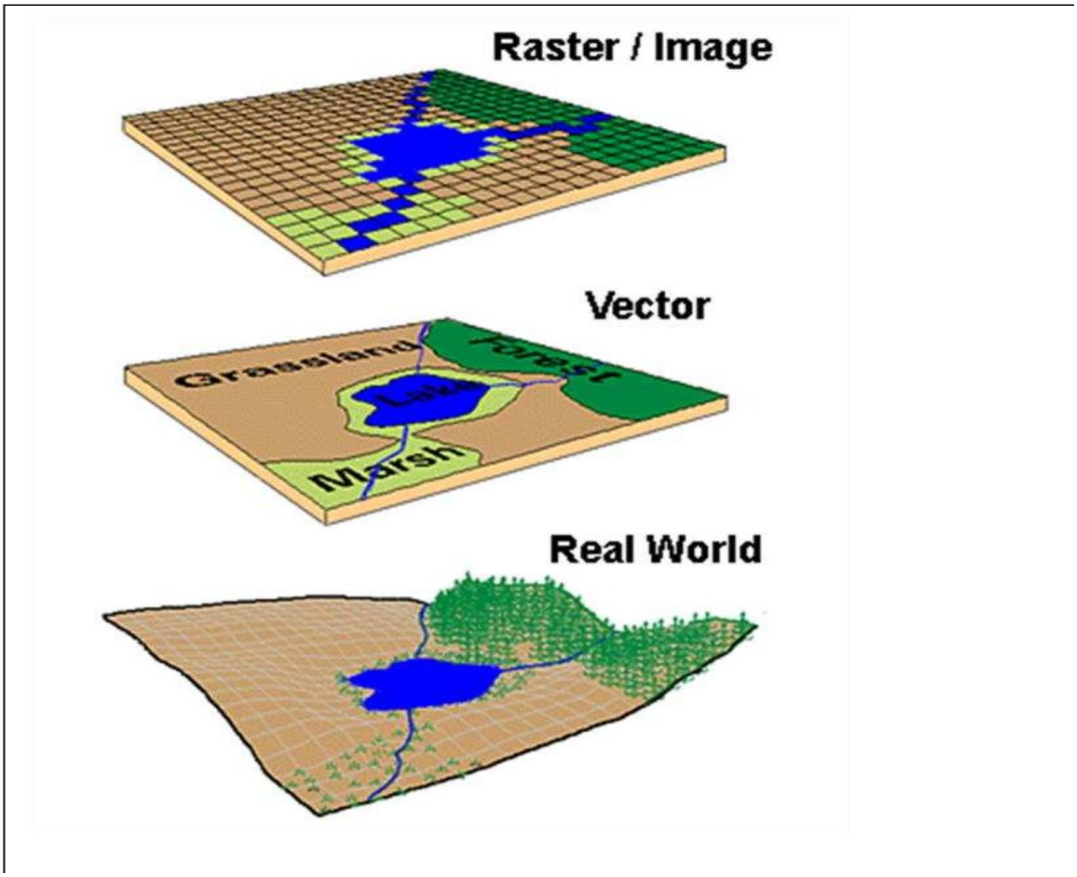
Relationships (rules or behavior): topological relationship such as coincidence, adjacency, and connectivity.

2.2 Spatial Data Models (Formats)

Traditionally spatial data has been stored and presented in the form of a map. Three basic types of spatial data models have evolved for storing geographic data digitally. These are referred to as:

- ❖ Vector
- ❖ Raster
- ❖ Image

The following diagram reflects the two primary spatial data encoding techniques. These are vector and raster.



a. Vector Model (Format)

Vector model uses discrete points, lines and/or areas corresponding to discrete objects with name or code number of attributes.

b. Raster Model (Format)

Raster model uses regularly spaced grid cells in specific sequence. An element of the grid cell is called a pixel which contains a single value of attributes.

c. Image data

utilizes techniques very similar to raster data, however typically lacks the internal formats required for analysis and modeling of the data. Images reflect *pictures* or *photographs* of the landscape.

The data model represents a set of guidelines to convert the real world (called entity) to the digitally and logically represented spatial objects consisting of the attributes and geometry.

2.3 GIS Data representations

Spatial data in GIS represents features that have a known location on the earth.

Points:	X & Y Locations
Polygon:	Connected X & Y Locations making a close figure.
Line:	Connected X & Y Locations without closing Row and column
Raster	Matrice represent geographic space.

2.4 Vector Data Structure

The method of representing geographic features by the basic graphical elements of **points, lines and polygon** is said to be the **vector method**, or **vector data model.** Vector data represent geographic space that is intuitive and reminiscent of analog maps.

Vector data mostly incorporate the use to **topological structure**, especially the ***Arc/Node structure*** explained below.

2.5 Rasters Data structure

Raster data models incorporate the use of a *grid-cell* data structure where the geographic area is divided into cells identified by row and column. A raster is a *tesselation* of a surface.• (A *tesselation* is defined as the process to cover a surface through the repeated use of a **single shape**.)

What is a raster data? A raster based system stores data by using a grid of cells. A unique reference coordinate represents each pixel either at a corner or in the middle of the cell. Each cell or pixel has discrete attributes assigned to it

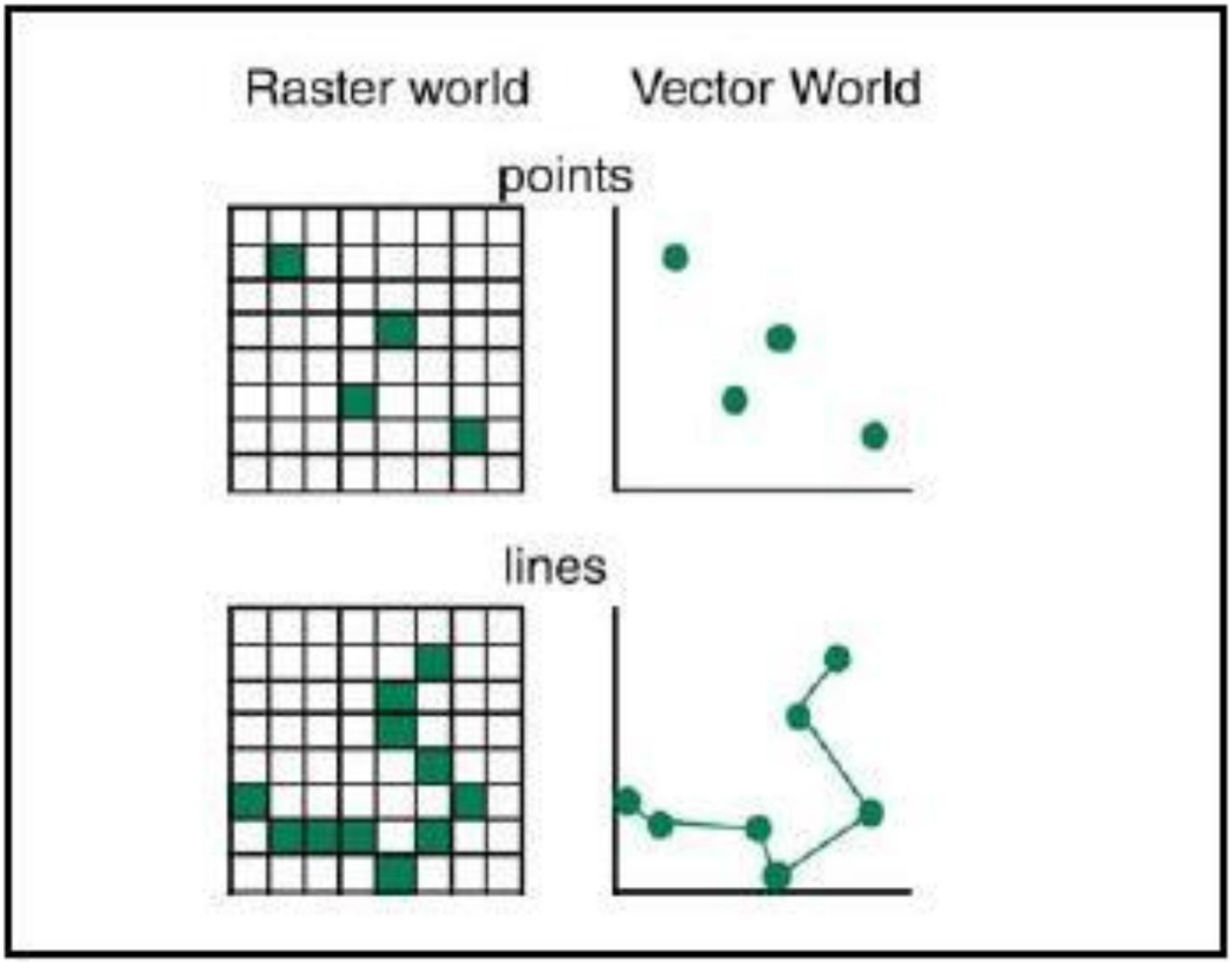
Raster data resolution is dependent on the pixel or grid size and may vary from sub-meter to many kilometers. raster data stores different information in layers; elevation, soil type, geology, forest type, rainfall rate, etc.

Generally, raster data requires less processing than vector data, but it consumes more computer storage space. Examples include:

- ❖ Remote sensors on satellites store data in raster format
- ❖ Digital terrain models (DTM) and digital elevation models (DEM)
- ❖ Continuous data (FIELD) suit a raster structure.

A raster can use any ***reasonable*** geometric shape, as long as it can be connected in such a way as to create a ***continuous*** surface.

Comparison Between Raster and Vector Data



Criterion	Raster Data	Vector Data
Representation	Continuous surfaces divided into cells (pixels).	Discrete features represented by points, lines, and polygons.
Examples	Satellite imagery, elevation models, temperature maps.	Roads, parcels, pipelines, administrative boundaries.
Geometry	Grid of cells, each with a value.	Exact coordinates forming shapes with attributes.
Resolution	Defined by cell size; accuracy limited by pixel resolution.	Defined by vertex precision; can achieve arbitrary accuracy.
File Size	Large (stores values for every cell, even empty areas).	Smaller (stores only coordinates and attributes).
Best Use Cases	Continuous phenomena (climate, terrain, population density).	Discrete features (infrastructure, property boundaries, networks).
Analysis Strengths	Map algebra, raster overlays, local neighborhood analysis.	Network analysis, spatial joins, buffering, topology queries.

Criterion	Raster Data	Vector Data
Formats	GeoTIFF, NetCDF, COG.	Shapefile, GeoJSON, GeoPackage.
Limitations	Editing individual features is difficult; precision limited by cell size.	Poor at modeling continuous phenomena; conversion to raster may be needed for environmental analysis.
Data Structure	It is simple data Structure.	It provides more compact data Structure than raster data
Operations	Overlay operations are easily and efficiently implement.	It provides efficiently encoding topology and as results more efficiently implementation of operations that requires topological information (such as network analysis)
Map	Is more or less required for efficient manipulation and enhancement of digital images.	Is better suited to supporting graphics that closely approximate hard-drawn maps.

2.6 Attribute Data and Attribute Data Model

Attribute data are the information linked to the geographic features (spatial data) that describe.

them. That is, attribute data are the "[n]on-graphic information associated with a point, line, or area elements in a GIS."

Attribute data model is a separate data model is used to store and maintain attribute data for GIS software. These data models may exist internally within the GIS software, or may be reflected in external commercial Database Management Software (DBMS). A variety of different data models exist for the storage and management of attribute data. The most common: tabular, hierarchical, network, relational and object-oriented models. Only relational models is commonly used in GIS currently.

2.7 Relational Model

The relational database organizes data in *tables*. Each table, is identified by a unique table name, and is organized by *rows* and *columns*. Each column within a table also has a unique name. Columns store the values for a specific attribute, e.g. cover group, tree height. Rows represent one record in the table. In a GIS each row is usually linked to a separate spatial feature, e.g. a forestry stand. Accordingly, each row would be comprised of several columns, each column containing a specific value for that geographic feature. The following figure presents a sample table for forest inventory features. This table has 4 rows and 5 columns. The forest stand number would be the *label* for the spatial feature as well as the *primary key* for the

database table. This serves as the linkage between the spatial definition of the feature and the attribute data for the feature.

UNIQUE STAND NUMBER	DOMINANT COVER GROUP	AVG. TREE HEIGHT	STAND SITE INDEX	STAND AGE
001	DEC	3	G	100
002	DEC-CON	4	M	80
003	DEC-CON	4	M	60
004	CON	4	G	120

Data is often stored in several tables. Tables can be joined or referenced to each other by common columns (relational fields). Usually, the common column is an identification number for a selected geographic feature, e.g. a forestry stand polygon number. This identification number acts as the *primary key* for the table. The ability to join tables through use of a common column is the essence of the relational model. Such relational joins are usually ad hoc in nature and form the basis of for querying in a relational GIS product.

The relational database model is the most widely accepted for managing the attributes of geographic data.

The relational DBMS is attractive because of its:

a) Simplicity in organization and data modelling

b) Flexibility - data can be manipulated in an ad hoc manner by joining tables.

c) Efficiency of storage - by the proper design of data tables redundant data can be minimized.

d) The non-procedural nature - queries on a relational database do not need to take into account the internal organization of the data.

The relational DBMS has emerged as the dominant commercial data management tool in GIS implementation and application.

2.8 Topology (Spatial Relationships)

The topologic model is often confusing to initial users of GIS. ***Topology is a mathematical approach that allows us to structure data based on the principles of feature adjacency and feature connectivity. It is in fact the mathematical method used to define spatial relationships.*** Without a topologic data structure in a vector based GIS most data manipulation and analysis functions would not be practical or feasible.

The most common topological data structure is the *arc/node* data model. This model contains two basic entities, the *arc* and the *node*.

- ❖ The arc is a series of points, joined by straight line segments, which start and end at a node. Arc presents **linear features**
- ❖ The node is an intersection point where two or more arcs meet. Nodes also occur at the end of a *dangling* arc, e.g. an arc that does not connect to another arc such as a dead end street.
- ❖ Isolated nodes, not connected to arcs represent **point features**.
- ❖ **A polygon feature** is comprised of a closed chain of arcs.

Topology is the "way in which geographical elements are linked together". Topology is how geographic features are related to one another and where they are in relation to one another.

Topology is the critical element that distinguishes a GIS from a graphics or automated cartography system. It is essential to the ability of a GIS to employ spatial relationships. Topology is what enables a GIS to emulate our human ability to discern and manipulate geographic relationships.

2.9 Labels

Affixed to data points, lines, or polygons.

- ❖ Used to describe the **feature** that you want to map.
- ❖ Can include text or numeric descriptors: i.e. nominal, ordinal, or interval/ratio data types.
- ❖ Must be careful in how the different data types are integrated and used – dangerous to mix and match.

2.10 GIS Database Creation/maintenance

Spatial data capture and edition goes hand in hand with GIS database creation. Sometime, or many a times, the database is designed long before data capture

Database is a large, computerized collection of structured data. In non-spatial domain, databases have been used since 1960s in bank account

administration, stock monitoring, salary administration, stores management, flight reservations etc.

Designing a database is not an easy task. One has to consider the purpose, available data sources and formats before data can be entered into the database. The format means data structure. Database should be properly maintained and up-to-date.

Database Management Systems (DBMS) are a software package that allows the user to set up, use and maintain a database.

GIS have inbuilt Ms Access for small databases, but has ArcSDE for large databases. GIS stores data mainly in relational database discussed above

Reasons for database

- ❖ DBMS supports storage and maintenance of very large volume of datasets.
- ❖ DBMS can be instructed to guard over data correctness or validity.
- ❖ DBMS supports concurrent use of the same dataset for many users.
- ❖ Provides high level declarative query language. A query is is a computer program that extracts data from the database based on specified condition.
- ❖ A DBMS supports use of data model. A data model is language which can apply to define data structure and in data manipulation and storage.

- ❖ Include data backup and recovery functions.
- ❖ Allows control over data redundancy.

Relational data model

A data model is a language that allows: The data structures to be stored; the integrity constraints, and the computer programme to manipulate the data.

For relational data model, the structure used to define the database is attributes, tuples and relations. Computer program can extract information applying query and change content (update)

In relational data model, a database is vided as a collection of relations, commonly known as tables. A table or a relation is a collection of tuples or records. An attribute is a named field of tuple. Same attribute comes from single domain (text/string, integer, real, date, etc)

When creating a relation is created, we need to indicate the type of tuple to strore:

- ❖ Provide name of relation.
- ❖ Indicate name of attributes.
- ❖ Set domain for each attribute.

Undelaying the attribute and their domain indicate primary key of the relations.

The set of tuples in a relation at a given point is called a relation stance.

Linking GIS and database

GIS software provide support for spatial data and attribute data. GIS stores spatial and attribute data separately, thus, it links them thorough a primary key.